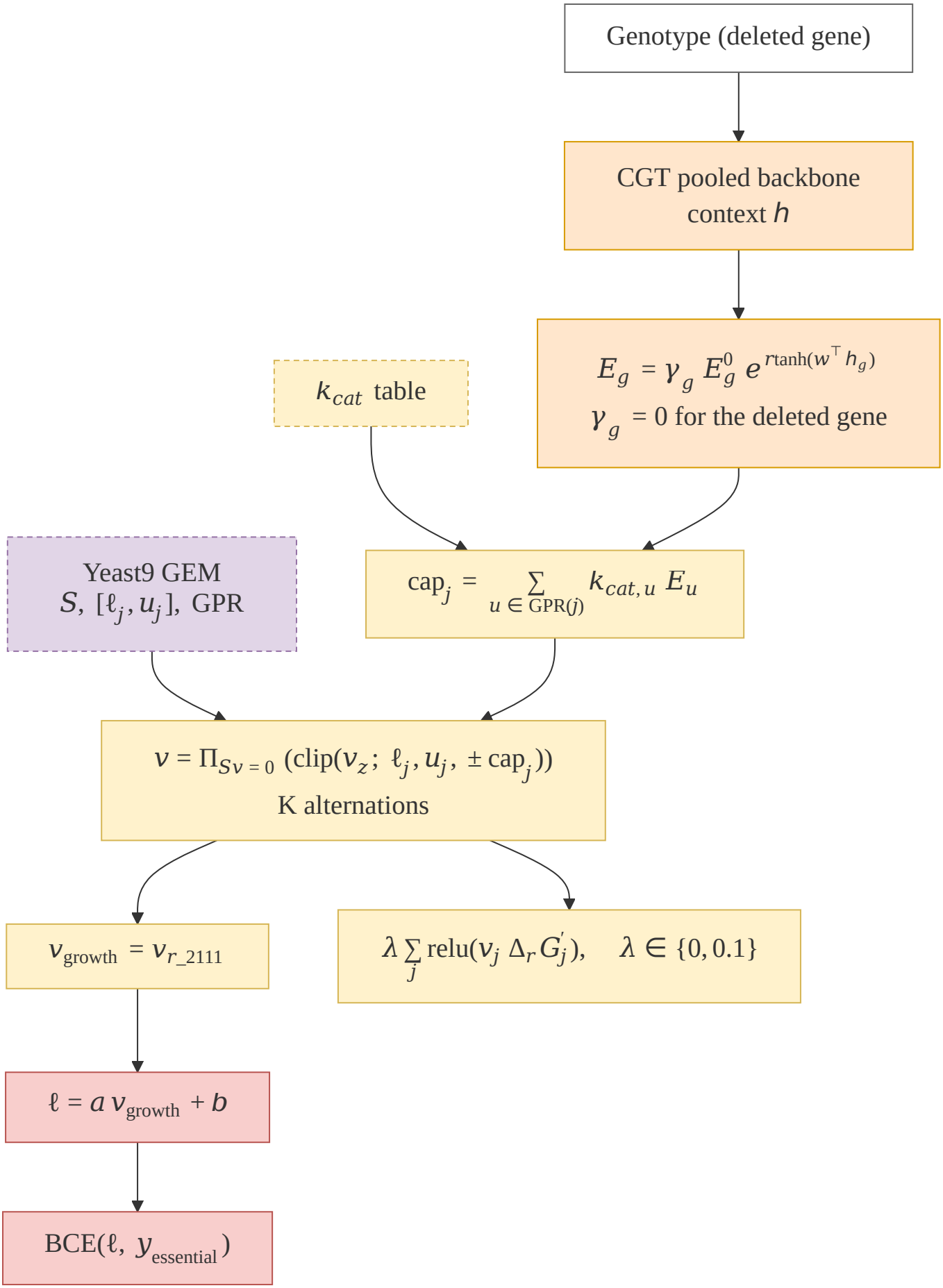

Genotype (deleted gene)
CGT pooled backbone context h
$E_g = \gamma_g E_g^0 e^{r \tanh(w^\top h_g)}$
$\gamma_g = 0$ for the deleted gene
k_{cat} table
Yeast9 GEM
$S,\ [\ell_j, u_j],$ GPR
$\text{cap}_j = \sum_{u \in \text{GPR}(j)} k_{cat,u} E_u$
$v = \Pi_{Sv=0}(\text{clip}(v_z;\ \ell_j, u_j, \pm \text{cap}_j))$
K alternations
$v_{\text{growth}} = v_{r_2111}$
$\lambda \sum_j \text{relu}(v_j \Delta_r G_j'), \quad \lambda \in \{0, 0.1\}$
$\ell = a\, v_{\text{growth}} + b$
$\text{BCE}(\ell,\ y_{\text{essential}})$